Follow the money: how fraud-linked Bitcoin transactions and wallets behave differently

Group name	**TODO:** group name
Dataset	Elliptic++ (Elmougy & Liu, KDD 2023): `github.com/git-disl/EllipticPlusPlus`
Code	**TODO:** github.com/<user>/<repo> (scripts `src/01--03`, README, results)

#	Name	Roll no.	Institute email	Contribution
1	**TODO:** name	**TODO:** roll no.	**TODO:** email	**TODO:** e.g. dataset, preparation, Tests 1–2 (1, 2, 3, 5)
2	**TODO:** name	**TODO:** roll no.	**TODO:** email	**TODO:** e.g. EDA, Tests 3–5, limitations, README (4, 5, 6)

1. Dataset

Source: Elliptic++, released by Georgia Tech with the KDD 2023 paper of Elmougy and Liu, which extends the Elliptic Bitcoin dataset of Weber et al. (2019). Downloaded on 21 October 2026 from the Google Drive folder linked in the repository above. Each row is a real Bitcoin transaction or wallet address; the blockchain analytics firm Elliptic labelled entities as *illicit* (scams, ransomware, malware, Ponzi schemes, dark markets), *licit* (exchanges, wallet providers, miners) or *unknown*. The data covers 49 time steps of about two weeks each.

What is raw about it: the labels sit in separate files from the features, coded 1/2/3; 77% of rows are *unknown*; 965 transactions miss all 17 named features; the wallet file repeats an address once for every time step it appears in; amounts and fees are extremely skewed; the money trail is only an edge list (who paid whom).

	Raw file	After encoding
Transactions (`txs_features` + `txs_classes`)		
Instances (rows)	203,769	202,804
Attributes (columns)	185	196
Instances × attributes	37,697,265	39,749,584
Wallets (`wallets_features` + `wallets_classes`)		
Instances (rows)	1,268,260	822,942
Attributes (columns)	58	66
Instances × attributes	73,559,080	54,314,172

Size check: transactions: $202{,}804 \times 196 = 39{,}749{,}584 \geq 3{,}000{,}000$; wallets: $822{,}942 \times 66 = 54{,}314{,}172 \geq 3{,}000{,}000$. Both tables meet the requirement on their own. Even the 17 named transaction features alone give $202{,}804 \times 17 = 3{,}447{,}668$.

Key features

Feature	Type	Real-world meaning
`class`	nominal	illicit / licit / unknown; our response
`num_output_addresses`	discrete	how many addresses receive money in the transaction
`num_input_addresses`	discrete	how many addresses pay into it
`fees`, `size`	continuous	fee paid to the miner (BTC); transaction size (bytes)
`fee_rate` (derived)	continuous	`fees`/`size` in satoshi per byte: the price paid for block space
`total_BTC`	continuous	total value moved (BTC)
`Time step`	discrete	1–49, about two weeks each
`num_txs_as_sender`, `_receiver`	discrete	a wallet's transactions sending / receiving money
`lifetime_in_blocks`	discrete	blocks between a wallet's first and last appearance

2. Data preparation

Issue found	What we did	Why, and not the alternative
Labels in separate files, coded 1/2/3	joined on `txId` / `address` (one-to-one, checked); mapped to names; one-hot encoded (`is_illicit`, `is_licit`, `is_unknown`)	a code like 3 is not a number: treating it as one would make "unknown" three times "illicit".
Unknown class (77% of transactions)	kept in the file, left out of every test	unknown is not "licit": counting it as licit would dilute the fraud signal with unlabelled fraud.
965 transactions miss all 17 named features	dropped (519 licit, 446 unknown, no illicit)	the 17 are missing *together*, a structural gap in the Elliptic++ augmentation, so imputing would invent whole transactions; no illicit row is lost.
Wallet file: 347,569 exact duplicate rows	dropped	they add no information and would count a wallet twice in every test.
61,487 wallets still listed in several time steps	collapsed to one row per address; first time step kept	we checked that every wallet feature is constant across these rows: they are whole-life summaries, so the repeats are the same wallet counted again.
Fee rate heavy-tailed (11,303 above the IQR fence, max 22,614 sat/B)	kept and flagged; tests on fee rate use $\log_{10}$	these are real, urgent payments; capping them would erase exactly the spread that Test 3 measures.
67 transactions with zero fee	kept; left out only where $\log_{10}$ fee rate is needed	a zero fee is real (miners accepted free transactions), but has no logarithm.
Money trail only an edge list	derived `receives_from_illicit`: the transaction spends an output of an illicit transaction	turns the graph into a feature that a test can use (Test 4).

We also checked Bitcoin's accounting identity, inputs $=$ outputs $+$ fee: it holds for 100% of rows, so amounts were not corrupted. There were no duplicate transactions.

3. What changed after preparation

Measure	Before	After
Transaction rows	203,769	202,804 (965 dropped)
Transaction attributes	185	196 (11 derived or encoded)
Missing cells (named features)	16,405	0
Wallet rows	1,268,260	822,942 (one per address)
Wallet duplicate rows	347,569	0
Wallets counted more than once	61,487	0
Wallet attributes	58	66 (pattern and one-hot class)
Fee-rate outliers	11,303	11,303, flagged and analysed on $\log_{10}$
Labelled transactions for tests	46,564	46,045 (4,545 illicit, 41,500 licit)
Labelled wallets for tests	265,354	265,354 (14,266 illicit, 251,088 licit)

The wallet table shrank by 35%, and no address was lost: all 822,942 remain, each counted once. On the log scale the fee rate is close to symmetric (skewness 0.27 for licit transactions), whereas the raw amounts span eight orders of magnitude (2.7×10^{-5} to 6,335 BTC). The encoded label and the new money-trail and fee-rate features are what make Tests 3 and 4 possible.

4. Basic questions and problem statement

Basic questions

1. *How common is fraud?* 9.9% of labelled transactions are illicit (4,545 of 46,045); 77% of all transactions carry no label.
2. *Does fraud move more money?* No: the median illicit transaction moves 0.12 BTC, the median licit one 0.48 BTC.
3. *Is fraud steady over time?* No: the illicit share of a time step ranges from 0.3% (step 46) to 36.5% (step 13) (Figure 1).

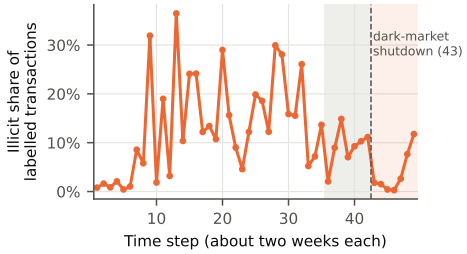


Figure 1: The illicit share swings from step to step and falls to near zero right after the dark-market shutdown at step 43 reported by Weber et al.; the shaded windows are those of Test 5.

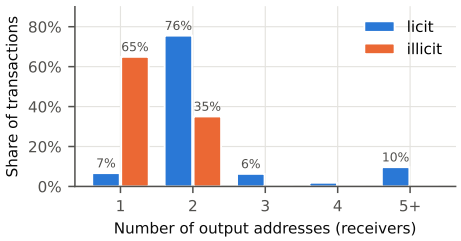


Figure 2: 65% of illicit transactions pay a single receiver; 76% of licit ones pay two, the typical "payment plus change".

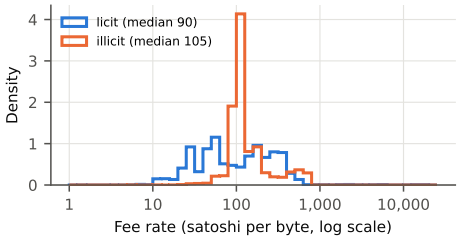


Figure 3: Illicit fee rates pile up in a narrow spike at about 100–120 sat/B; licit fee rates spread over two orders of magnitude.

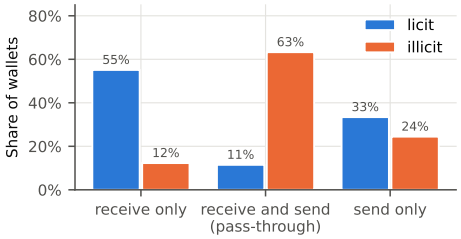


Figure 4: 63% of illicit wallets both receive and send money (pass-through), against 11% of licit wallets, most of which only receive.

Our problem statements

P1 (primary). Does illicit money move in a different *shape* from legitimate money: forwarded whole to a single receiver, through wallets that pass it straight on? (Tests 1, 2)

P2. Does fraud leave a *fingerprint* in the fees: are illicit fee rates more uniform, as automated software with a fixed fee would produce? (Test 3)

P3. Does fraud *follow the money trail*: is a transaction that spends illicit money itself more likely to be illicit, comparing like with like in time? (Test 4)

P4. Did the dark-market shutdown at step 43 lower the illicit share? (Test 5)

How they are novel: the basic questions compare totals and averages, one row at a time. Ours ask about *behaviour*: how money is split (outputs), how a wallet uses it (receive / send), how *variable* the fees are rather than how high, and what the *previous* transaction on the money trail was, which no single row records and has to be derived from the graph. Elliptic++ has so far been used mainly to train classifiers; we test these behaviours directly, with effect sizes.

5. Statistical inferences

Rows tested: only labelled rows: 46,045 transactions (4,545 illicit, 41,500 licit) and 265,354 wallets (14,266 illicit, 251,088 licit). All tests use $\alpha = 0.05$. Hypotheses, sides and the smallest differences that matter were fixed in the code before any test statistic was computed. Where a p-value is below the smallest number a computer can store ($\approx 10^{-308}$), we give its order of magnitude from the log p-value.

Test 1 (P1): do illicit transactions pay a single receiver more often?

a) Hypotheses, written before running the test

H_0	$p_{\text{ill}} - p_{\text{lic}} \leq 0$
H_1	$p_{\text{ill}} - p_{\text{lic}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test (pooled)
Smallest difference that matters	10 percentage points

p: share of transactions with exactly one output address. One-sided, because P1 claims illicit money is forwarded whole. A transaction with one receiver and no change output is the signature of "sweeping" an address empty; 10 points is the gap at which an analyst screening transactions would gain from the rule.

b) Other tests we considered

The response is binary, the two groups are independent and large (the smallest expected cell count is 564), so the normal approximation of the z-test holds, and it gives the difference in proportions directly. With x single-output transactions out of n in each class, $\hat{p} = x/n$:

$$z = \frac{\hat{p}_{\text{ill}} - \hat{p}_{\text{lic}}}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_{\text{ill}}} + \frac{1}{n_{\text{lic}}}\right)}}, \qquad \bar{p} = \frac{x_{\text{ill}} + x_{\text{lic}}}{n_{\text{ill}} + n_{\text{lic}}},$$

$p = 1 - \Phi(z)$. The Newcombe interval for $d = \hat{p}_{\text{ill}} - \hat{p}_{\text{lic}}$ combines the Wilson score intervals (l_i, u_i) of the two proportions:

$$L = d - \sqrt{(\hat{p}_{\text{ill}} - l_{\text{ill}})^2 + (u_{\text{lic}} - \hat{p}_{\text{lic}})^2},$$

$$U = d + \sqrt{(u_{\text{ill}} - \hat{p}_{\text{ill}})^2 + (\hat{p}_{\text{lic}} - l_{\text{lic}})^2}.$$

Also considered	Why we did not use it
Pearson χ^2 (2×2)	the same test squared ($\chi^2 = 12{,}780 = z^2$, d.f. 1), but two-sided only.
Fisher's exact test	exact, but built for small counts; here it agrees (OR $= 25.9$, $p < 10^{-300}$) and adds nothing.
Welch t on `num_output_addresses`	compares means of a count driven by licit batch payouts with up to 13,107 outputs (mean 6.17 vs 1.35); it answers a different question.

c) Result

Estimate	$+58.2$ points (64.9% illicit, 6.7% licit)
95% confidence interval (Newcombe)	$+56.8$ to $+59.6$ points
Test statistic (d.f.)	$z = 113.0$ (none: normal reference)
p-value	$\approx 10^{-2778}$
Decision	reject H_0

Illicit transactions pay a single receiver nearly ten times as often as licit ones. The whole interval lies far beyond 10 points, so the difference matters. Together with the 89% of illicit transactions that have a single input (licit: 50%), the typical illicit transaction is "one address in, one address out".

Test 2 (P1): do illicit wallets pass money straight through?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\pi_{\text{ill},k} = \pi_{\text{lic},k}$ for every pattern k (independence)
$\boldsymbol{H_1}$	$\pi_{\text{ill},k} \neq \pi_{\text{lic},k}$ for at least one k
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2 (any departure)
Test	Pearson χ^2 test of independence, 2×3
Smallest difference that matters	Cramér's $V \geq 0.10$ and a 10-point gap in the pass-through share

$\pi_{c,k}$: share of class-c wallets with pattern k: *receive only*, *pass-through* (receives and sends) or *send only* (received before the data window). $V = 0.10$ is the conventional "small" association; we add a 10-point gap so that the effect can be read in wallets.

b) Other tests we considered

Two nominal variables, one with three levels, and independent wallets: this is the setting of the χ^2 test of independence. With observed counts O_{ik} (class i, pattern k), row totals R_i, column totals C_k and n wallets:

$$\chi^2 = \sum_{i=1}^{2}\sum_{k=1}^{3} \frac{(O_{ik} - E_{ik})^2}{E_{ik}}, \qquad E_{ik} = \frac{R_i C_k}{n},$$

$$\nu = (2-1)(3-1) = 2, \qquad V = \sqrt{\frac{\chi^2}{n\,(\min(2,3)-1)}},$$

$$r_{ik} = \frac{O_{ik} - E_{ik}}{\sqrt{E_{ik}}}.$$

r_{ik} is the Pearson residual of each cell. The smallest expected count is 2,035, far above 5.

Also considered	Why we did not use it
G-test (likelihood ratio)	asymptotically equivalent; same decision ($G = 21{,}428$, d.f. 2). Pearson's is the standard to report.
Two-proportion z on pass-through only	throws away the other two patterns; it agrees ($z = 172.1$). We use it for the estimate below.
Mann–Whitney on `total_txs`	tests whether one class has more transactions, not *how* wallets use money.

c) Result

Estimate	pass-through share $+51.8$ points (63.3% vs 11.5%); Cramér's $V = 0.338$
95% confidence interval	$+51.0$ to $+52.6$ points; V: 0.334 to 0.343 (bootstrap)
Test statistic (d.f.)	$\chi^2 = 30{,}397$ (2)
p-value	$\approx 10^{-6601}$
Decision	reject H_0

The association is moderate ($V \approx 0.34$) and both intervals clear our thresholds. The Pearson residuals r_{ik} show where it comes from: illicit pass-through $+155$, illicit receive-only -67. Fraud-linked wallets act as relays, not as savings.

Test 3 (P2): are illicit fee rates less variable?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\sigma_{\text{ill}} \geq \sigma_{\text{lic}}$
$\boldsymbol{H_1}$	$\sigma_{\text{ill}} < \sigma_{\text{lic}}$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Brown–Forsythe test
Smallest difference that matters	SD ratio $\sigma_{\text{ill}}/\sigma_{\text{lic}} \leq 0.80$

σ: standard deviation of $\log_{10}$ fee rate (4,545 illicit, 41,452 licit; 48 zero-fee licit rows excluded). One-sided, because P2 claims uniformity. A spread 20% narrower is what a monitoring rule could pick up.

b) Other tests we considered

We compare spreads, not means. Brown–Forsythe (Levene with the median) runs a t-test on the absolute deviations from each group's median, so it does not need normality and gives a one-sided p-value. For log fee rate x_{ij} of transaction i in group j, with group median $\tilde{x}_j$:

$$z_{ij} = |x_{ij} - \tilde{x}_j|, \qquad t = \frac{\bar{z}_{\text{ill}} - \bar{z}_{\text{lic}}}{s_p\sqrt{\frac{1}{n_{\text{ill}}} + \frac{1}{n_{\text{lic}}}}}, \qquad \nu = n_{\text{ill}} + n_{\text{lic}} - 2,$$

where s_p^2 is the pooled variance of the z_{ij}, and $F = t^2$ is Levene's statistic. The estimate $s_{\text{ill}}/s_{\text{lic}}$ gets a percentile bootstrap interval: the 2.5th and 97.5th percentiles of the ratio over 2,000 resamples drawn within each group.

Also considered	Why we did not use it
F-test of variances	valid only for normal data; illicit log rates are skewed (1.22) and Shapiro–Wilk rejects normality ($p \approx 10^{-59}$). It agrees ($F = 0.31$).
Bartlett's test	just as sensitive to non-normality ($\chi^2 = 2{,}051$).
Fligner–Killeen	rank-based and also valid; agrees ($\chi^2 = 2{,}538$), but two-sided only.
Wilcoxon signed-rank, by time step	controls for the fee market changing over time; we run it as a check (below).

c) Result

Estimate	SD ratio 0.559 (0.255 vs 0.456 in $\log_{10}$)
95% confidence interval	0.543 to 0.575 (bootstrap, 2,000 resamples)
Test statistic (d.f.)	$t = -57.87$ (45,995); $F = t^2 = 3{,}349$
p-value	$\approx 10^{-729}$
Decision	reject H_0

Illicit fee rates are about 44% less spread out, and the whole interval is below 0.80. The effect is not a by-product of time: within the same time step, illicit fees are the narrower group in 41 of 45 steps (median SD 0.082 vs 0.206; Wilcoxon $W = 106$, $p = 2.3 \times 10^{-7}$). The fingerprint is concrete: 25.4% of illicit transactions pay a fee of exactly 0.0002 BTC (licit: 8.7%), typically on a 192-byte, one-in-one-out transaction, which gives the spike at 104 sat/B in Figure 3.

Test 4 (P3): does fraud follow the money trail?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	common odds ratio $\mathrm{OR}_{\mathrm{MH}} \leq 1$
$\boldsymbol{H_1}$	$\mathrm{OR}_{\mathrm{MH}} > 1$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Cochran–Mantel–Haenszel (CMH), stratified by time step
Smallest difference that matters	OR ≥ 2

Exposure: the transaction spends an output of an illicit transaction. Outcome: it is illicit. We stratify because a busy fraud period (Figure 1) makes both the exposure and the outcome common at once, which alone would create an association. An OR of 2 doubles the odds, the level at which flagging a payer's payees is worth an analyst's time.

b) Other tests we considered

A 2×2 table in each of 48 time steps, independent transactions: CMH tests a common odds ratio across strata while comparing like with like in time. In time step k, a_k = exposed illicit, b_k = exposed licit, c_k = unexposed illicit, d_k = unexposed licit, $n_k = a_k + b_k + c_k + d_k$:

$$\chi^2_{\mathrm{CMH}} = \frac{\left[\sum_k \left(a_k - \mathrm{E}(a_k)\right)\right]^2}{\sum_k \mathrm{Var}(a_k)}, \qquad \mathrm{OR}_{\mathrm{MH}} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k},$$

$$\mathrm{E}(a_k) = \frac{(a_k + b_k)(a_k + c_k)}{n_k},$$

$$\mathrm{Var}(a_k) = \frac{(a_k + b_k)(c_k + d_k)(a_k + c_k)(b_k + d_k)}{n_k^2 (n_k - 1)}.$$

$\nu = 1$; the one-sided p is $1 - \Phi\left(\mathrm{sign}(\mathrm{OR}_{\mathrm{MH}} - 1)\sqrt{\chi^2_{\mathrm{CMH}}}\right)$. The interval uses the Robins–Breslow–Greenland variance of $\ln \mathrm{OR}_{\mathrm{MH}}$.

Also considered	**Why we did not use it**
Pearson χ^2 / Fisher, pooled	ignore time and so are confounded: crude OR 31.9 vs 19.3 stratified.
Logistic regression with step dummies	agrees in direction (OR 24.7, 21.4 to 28.6), but needs a model with 49 parameters; CMH is the direct test.
Breslow–Day (homogeneity)	not a test of H_0 but of CMH's assumption; run below.

c) Result

Estimate	$\mathrm{OR}_{\mathrm{MH}} = 19.3$ (73.8% vs 8.1% illicit)
95% confidence interval	16.7 to 22.3
Test statistic (d.f.)	$\chi^2_{\mathrm{CMH}} = 2{,}906$ (1)
p-value	$\approx 10^{-633}$
Decision	reject H_0

Even within the same two weeks, a transaction that spends illicit money has 19 times the odds of being illicit itself; risk ratio 9.1 (8.7 to 9.5). Breslow–Day rejects a common OR ($\chi^2 = 235$, 38 d.f., $p \approx 10^{-29}$): the size of the effect varies over time. Its direction does not: the odds ratio is at least 2 in 46 of the 48 time steps (median 15.3), so the conclusion stands.

Test 5 (P4): did the illicit share fall after the dark-market shutdown?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_{\text{after}} - \mu_{\text{before}} \geq 0$
$\boldsymbol{H_1}$	$\mu_{\text{after}} - \mu_{\text{before}} < 0$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Welch t-test on per-step illicit shares
Smallest difference that matters	5 percentage points

μ: mean illicit share per time step, steps 36–42 (before) and 43–49 (after), 7 each. One-sided, because a shutdown should remove fraud. 5 points is about half the pre-shutdown level, a drop that would show in an exchange's compliance workload.

b) Other tests we considered

The unit is the time step, not the transaction: transactions in one step share a market, so pooling them would overstate the evidence. With 7 values per group we need a small-sample test that does not assume equal variances. With group means $\bar{x}$, variances s^2 and sizes n (a = after, b = before):

$$t = \frac{\bar{x}_a - \bar{x}_b}{\sqrt{s_a^2/n_a + s_b^2/n_b}}, \qquad \nu = \frac{\left(s_a^2/n_a + s_b^2/n_b\right)^2}{\frac{(s_a^2/n_a)^2}{n_a - 1} + \frac{(s_b^2/n_b)^2}{n_b - 1}},$$

(Welch–Satterthwaite), and the interval is $(\bar{x}_a - \bar{x}_b) \pm t_{0.975,\nu}\sqrt{s_a^2/n_a + s_b^2/n_b}$.

Also considered	Why we did not use it
Pooled two-proportion z on 15,014 transactions	treats transactions as independent: $z = -15.7$, $p = 7 \times 10^{-56}$. Far too confident.
Student t	variances look equal (Levene $p = 0.90$), giving the same t; Welch was fixed in advance.
Mann–Whitney (exact)	robust to the non-normal "after" group (Shapiro $p = 0.04$); agrees: $U = 9, p = 0.027$.
Exact permutation test	all 3,432 splits; agrees: $p = 0.020$.

c) Result

Estimate	-5.4 points (3.7% after, 9.1% before)
95% confidence interval	-10.2 to -0.5 points
Test statistic (d.f.)	$t = -2.43$ (11.9)
p-value	0.016
Decision	reject H_0

The illicit share did fall after step 43, and all four tests agree. Whether the fall *matters* is not settled: the estimate just passes our 5 points, but the interval reaches -0.5, so a trivial drop cannot be ruled out with 7 steps. The shares at steps 48 and 49 (7.6%, 11.8%) show fraud returning, so the effect looks temporary.

6. Limitations

- **Labels come from one firm.** "Illicit" is Elliptic's judgement, from its own heuristics; if those heuristics follow the money trail, Test 4 partly measures the labelling. We cannot check this from the data.
- **77% of transactions are unknown.** Our results hold for labelled transactions, which may be the easier cases to label.
- **Association, not cause.** We show how fraud-linked money behaves, not that the behaviour causes or proves fraud; licit services (exchanges sweeping deposits) also forward money whole.
- **Huge samples.** With 46,045 rows almost any difference is significant, which is why we judge each result by its interval against a pre-set threshold.
- **Shutdown dating.** Step 43 is the date reported by Weber et al.; we did not date it independently, and with 7 steps per side Test 5 has low power.
- **One period.** The data covers 49 two-week steps of Bitcoin; fee markets and fraud methods have changed since.

References. M. Weber et al., "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics", KDD Workshop on Anomaly Detection in Finance, 2019. Y. Elmougy and L. Liu, "Demystifying Fraudulent Transactions and Illicit Nodes in the Bitcoin Network for Financial Forensics", KDD 2023.